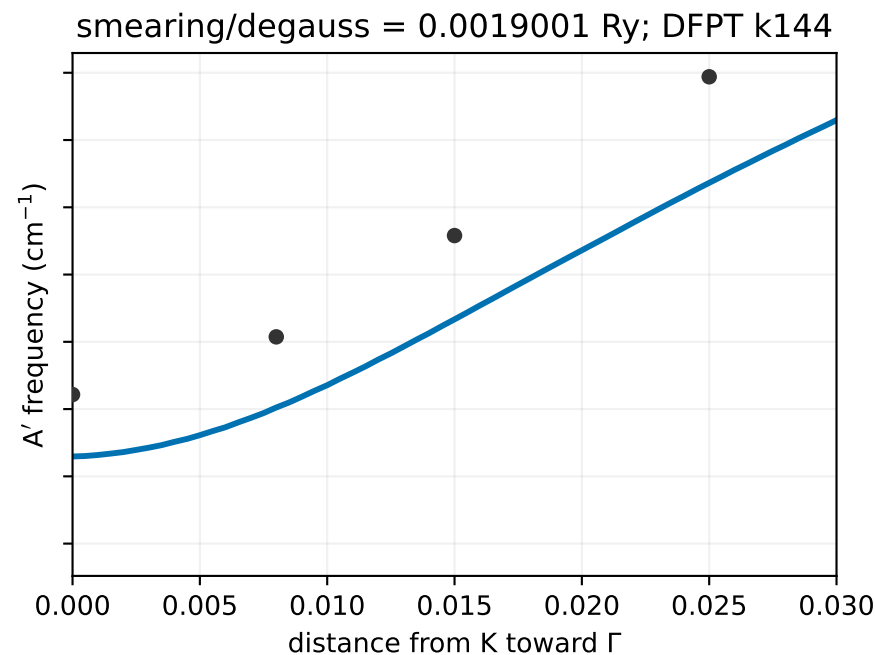
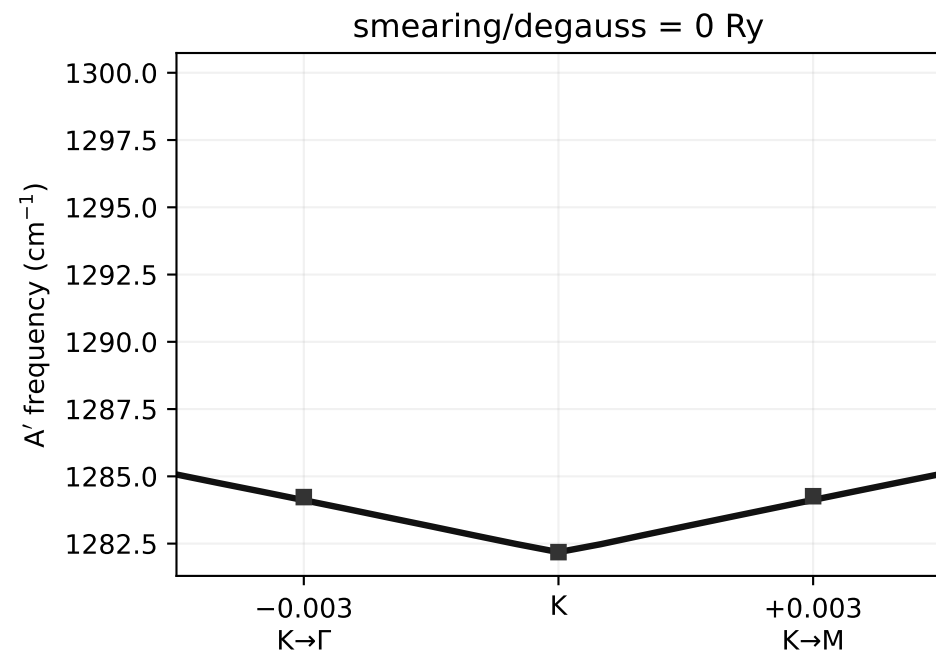
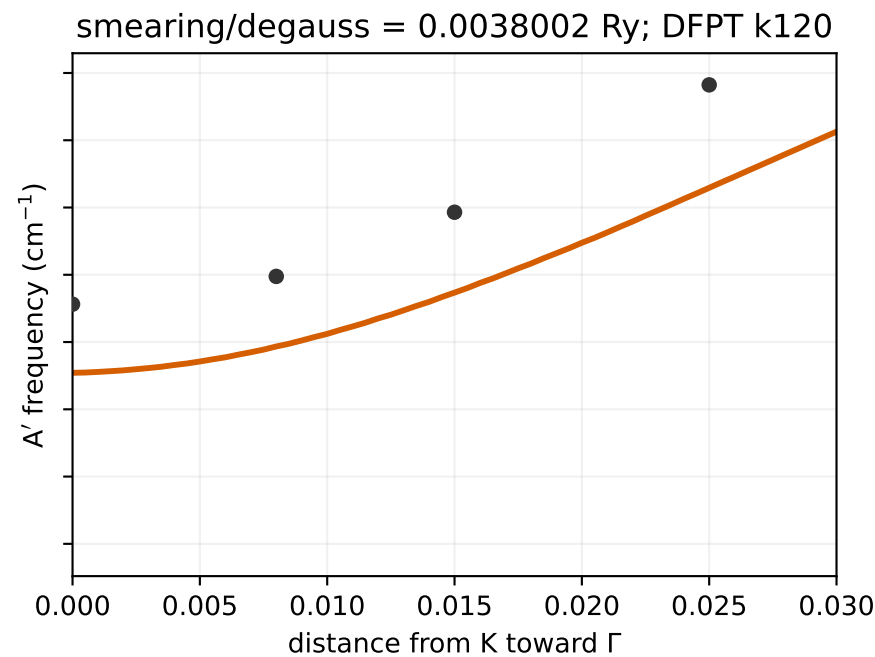
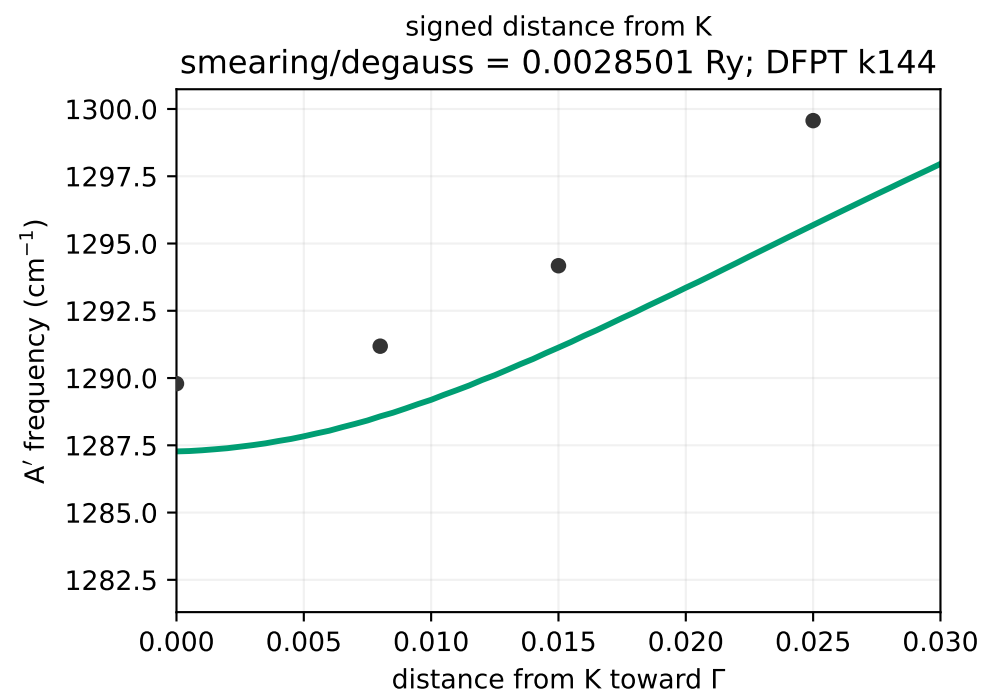


Graphene A': fixed-background MLIP + full EPC LR versus DFPT



- MLIP + full EPC LR
- direct DFPT
- zero-smearing DFPT $1/N_k$ extrapolation



Finite-smearing $K \rightarrow \Gamma$ absolute errors:
0.0019001: RMSE 3.06, K offset -2.30 cm^{-1}
0.0028501: RMSE 3.06, K offset -2.52 cm^{-1}
0.0038002: RMSE 3.04, K offset -2.55 cm^{-1}

Zero-smearing $d=0.003$:
quadrature-extrapolated depth errors
 $K \rightarrow \Gamma$ 3.94%; $K \rightarrow M$ 4.92%

Finite DFPT is available only on
the common $K \rightarrow \Gamma$ direction.